

COURSE GLOSSARY

Unsupervised Learning in Python

Centroid: The mean (average) point of all samples in a cluster, used by k-means to represent and assign membership to clusters

Clustering: An unsupervised learning task that groups samples so that items in the same group are more similar to each other than to those in other groups

Cosine similarity: A measure of similarity between two vectors based on the cosine of the angle between them, where values closer to 1 indicate more similar direction (topic mixture) irrespective of magnitude

Cross-tabulation (crosstab): A contingency table that shows the frequency counts of combinations of two categorical variables, commonly used to compare cluster labels with known class labels

csr_matrix (sparse matrix): A compressed sparse row format that efficiently stores only the non-zero entries of a large sparse array, saving memory and enabling operations on sparse datasets

Dendrogram: A tree-like diagram that visualizes the sequence of merges or splits produced by hierarchical clustering and encodes the distances at which clusters are joined

Explained variance: The amount of variance captured by each principal component, used to assess how many components are needed to approximate the original data (intrinsic dimension)

Hierarchical clustering: A clustering approach that builds a tree of clusters by either merging (agglomerative) or splitting (divisive) clusters to represent nested groupings at multiple scales

Inertia: A measure used in k-means equal to the sum of squared distances between samples and their assigned centroids, where lower values indicate tighter (less spread out) clusters

Intrinsic dimension: The minimal number of features or directions required to approximate a dataset well, often estimated by counting PCA components with significant explained variance

k-means clustering: A popular clustering algorithm that partitions samples into a specified number k of clusters by iteratively assigning samples to the nearest centroid and updating centroids to be cluster means

Learning rate (t-SNE): A hyperparameter controlling the optimization step size in t-SNE, where poor choices can cause points to collapse or be poorly separated and typically values between ~50 and 200 are tried

Linkage method: A rule that defines the distance between clusters in hierarchical clustering (e.g., complete linkage uses the maximum pairwise distance between cluster members)

Non-negative Matrix Factorization (NMF): A parts-based dimension-reduction technique that decomposes non-negative data into non-negative components and feature coefficients, producing interpretable additive parts or topics

Pipeline (make_pipeline): A scikit-learn utility that chains preprocessing and modeling steps so data flows automatically from one step to the next during fit and predict/transform operations

Principal Component Analysis (PCA): A linear dimension-reduction method that rotates and recenters data to new orthogonal axes (principal components) ordered by decreasing variance, enabling compression and decorrelation

Principal components: The orthogonal directions (vectors) learned by PCA that capture the axes of maximal variance in the data and are available as the components attribute after fitting

StandardScaler: A preprocessing transformer that standardizes features by removing the mean and scaling to unit variance so each feature has mean 0 and variance 1

t-SNE (t-distributed stochastic neighbor embedding): A nonlinear dimension-reduction technique that maps high-dimensional data into 2 or 3 dimensions to preserve local neighbor relationships for visualization

tf-idf: A text weighting scheme combining term frequency (tf) and inverse document frequency (idf) to quantify the importance of a word in a document relative to a corpus

TruncatedSVD: A linear dimension-reduction method equivalent to PCA for dense data but designed to work with sparse matrices (e.g., `csr_matrix`), commonly used on word-frequency data

Unsupervised learning: A class of machine learning methods that discover patterns or structure in data without using labeled outcomes, such as clustering or dimension reduction